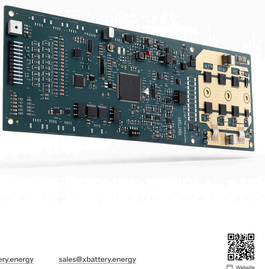
XBATTERY

BharatBMS ESS



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Website

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Xbattery BharatBMS-ESS

The BharatBMS-ESS redefines energy management with intelligent performance, robust safety, and scalable design. Engineered for next-generation ESS solutions, it supports up to 4ZS LFP and NMC battery packs, ensuring precise monitoring, intelligent cell balancing, and multi-level protection, all in a compact, space-efficient form.

Applications

Backup & UPS Systems
Uninterrupted, stable energy support

Telecom Infrastructure
Reliable backup for towers

Industrial Battery Packs
High-performance power systems

Renewable Energy Storage
Solar and wind power systems

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Core Features

Intelligent Cell Monitoring
Supports up to 42 series cells with ±2 mV precision and passive balancing.

Communication & Control
CAN FD, UART, SPI, and Ethernet for real-time data and remote monitoring.

Safety First
ISO 26262-compliant protection with fault diagnostics for voltage, current, and temperature.

Thermal Management
Real-time temperature sensing and thermal runaway detection for efficient heat control.

Scalable Architecture
Modular design allows flexible series and parallel configurations.

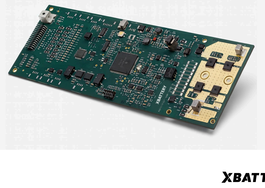
SOC/SOH Algorithms
Accurate State-of-Charge and State-of-Health estimation.

Additional Features

Smart Diagnostics
Real-time visualization and lifecycle analytics for optimal performance.

Optimized Software
High-efficiency C framework for real-time applications.

Customizable API
Open API for seamless integration and control.



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ESS BMS Offerings

01
Xbattery BharatBMS-ESS-48V
Smart BMS For UPS, Telecom, And Small ESS Applications, Supports Up To 28S For Compact And Reliable Energy Storage.

02
Xbattery BharatBMS-ESS-72V
High-Efficiency BMS For Industrial And Medium-Scale ESS, Supports Up To 38S For Safe And Efficient Power Management.

03
Xbattery BharatBMS-ESS-110V
Advanced BMS For Large ESS And Grid Storage, Supports Up To 42S For High-Voltage, High-Performance Energy Systems.

Product Specifications

Specification	ESS 48V	ESS 72V	ESS 110V
Cell Configuration	Up to 28S	Up to 28S	Up to 42S
Nominal Pack Voltage Range	48.0V	72.0V	110.0V
Max Pack Voltage Range	100.8V	100.8V	151.2V
Min Pack Voltage Range	70.0V	70.0V	105.0V
Typical Applications	UPS, Telecom, Small ESS	Industrial ESS, Medium Storage	Large ESS
Battery Capacity Range	25kWh	25kWh	38kWh

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Tech Specifications

General & Electrical Specifications	Parameters
Input Power & Communication Power Requirements	9V–60V input, 12V typical. CAN-powered or separate DC
Cell Voltage & Balancing	
Cell Voltage Measurement Range	1.5V to 4.2V
Measurement Accuracy & Resolution	±5mV, 1mV resolution
Balancing Topology (active or passive/dissipative)	Passive
Balancing Current	~30 to 100mA (depending on configuration)
Balance Enable/Threshold Voltages	Typically enabled >3.45V
Temperature Sensing & Thermal Management	
Number & Type of Temperature Sensors Supported	4–8 NTCs (10kΩ typical)
Sensor Range & Accuracy	±1°C
Thermal Protection Thresholds and cut-offs	-20°C to +60°C (pack operation), >0°C or >60°C triggers cutoff
Current Handling	
Continuous Charge Current Limit	100A (based on 100Ah pack, 1C)
Peak Charge & Discharge Currents (duration and recovery conditions)	150A for >10s
Continuous Discharge and Peak Discharge Limits	100A/150A (>10 s)
Internal Circuit Resistance	<5mΩ (BMU + cabling)
Standards & Certifications	
Safety Standards	AIS-038 Rev 2, IEC 62133-2
EMS/EMC compliance, regulatory notes	BIS applicable, CISPR 25 in roadmap
Performance & Lifecycle	
Efficiency	>98%
Power Consumption (active/idle/sleep current)	200mW/50mW/<5mW
Cycle Life Expectations (if gauging capacity)	2000+ cycles (depends on cell)

Mechanical & Environmental	Parameters
Dimensions & Weight	Custom board, ~100mm × 70mm (est.)
Mounting Form Factor/Connector Types	Configurable
Operating & Storage Temperature Range	-20°C to +60°C , -40°C to +70°C
Humidity/Environmental Classifications (indoor/outdoor, shock & vibration specs)	10–90% RH, non-condensing
Ingress/IP rating (if enclosure provided)	IP20–IP65 (depending on enclosure
Software, Diagnostics & Tools	
Configuration Software Tools	PC-based GUI or CLI over UART/CAN
Event/Error Logging & RTC capabilities	Timestamped fault log (non-volatile) ; Yes (battery-backed optional)
Diagnostics & Self-Test features	Self-test on boot, watchdog, CAN alive
Versioning & Firmware upgrade method	UART or CAN bootloader supported
Protections & Safety	
Over-voltage Protection (per cell and pack)	Cell: >3.65V; Pack: >58.8V
Under-voltage Protection (per cell and pack)	Cell: <2.5V; Pack: <36V
Over & Under-temperature cut-offs	Charge: 0°C to 45°C; Discharge: -20°C to 60°C
Leakage/Ground fault detection	Optional (RCD/GFD integration)
Protection switching topology	MOSFET/Contactor based
Precharge circuit or inrush current limiting	Yes (resistive soft start/inrush limiter)
Communication Interfaces	
Supported Protocols	CAN 2.0B (500kbps), UART (optional), BLE (ESP8266)
Baud Rates & CAN Speeds	CAN: 125k to 1 Mbps; UART: 9600–115200
General-purpose I/O	6 (Analog inputs)
Host Interfaces	CAN or UART to host/controller/charger